DATA SCIENCE MINOR PROJECT REPORT

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Other Conventions

i. **Please note the case of letters in the cover page:** The 3rd line is 16 pt bold and other lines are 12 pt. The page is centred. Department and Institute names are bold.
ii. All the matter contained in the report should be typed in MS word (1.5 spacing) Times New Roman, 12 pt or equivalent with other software.
iii. Figures and tables may be inserted in the text as they appear or may be appended in order.
iv. Table of Content shall be in well hyperlinked
v. List of figures and tables shall be maintained with captions in MS word.
vi. List of references shall be appended at the end.
vii. References shall be in IEEE format
viii. Total Number of pages with A4 size paper shall be minimum 30 pages and maximum 80 pages.

COVER PAGE

DATA ANALYSIS AND VISUALIZATION

PROJECT REPORT

(Project Semester January-April 2026)

Consumer Shopping Trends Analysis

Submitted by:-

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Registration No.: 12415508

Programme and Section: B.Tech CSE and 224YM

Course Code: INT387

Under the Guidance of

Dr. Baljinder Kaur **(UID:-27952)**

Discipline of CSE/IT

Lovely School of _____

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that **ARUNAV KUMAR** bearing Registration no. **12415508** has completed INT387 project titled, **"CONSUMER SHOPPING TRENDS ANALYSIS"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 22/04/2026

DECLARATION

I**, ARUNAV KUMAR**, student of **Computer Science and Engineering** under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 22/04/2026 Signature

Registration No.: 12415508 Name: Arunav Kumar

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1) INTRODUCTION

In the modern digital economy, consumer behaviour has become increasingly dynamic and complex due to the rapid growth of online platforms, technological advancements, and changing lifestyle patterns. Businesses today rely heavily on data-driven insights to understand customer preferences, improve decision-making, and stay competitive in the market. Analysing shopping trends plays a vital role in identifying purchasing patterns, customer segmentation, and market demands.

This project, titled **"Consumer Shopping Trends Analysis 2026"**, focuses on examining consumer purchasing behaviour using a structured dataset named *Consumer_shopping_Trends_2026*. The dataset includes various attributes such as customer demographics, product categories, purchase frequency, payment methods, and transaction amounts. By analysing these variables, the project aims to uncover meaningful insights that can help businesses enhance their strategies and improve customer satisfaction.

The analysis and visualization of the dataset have been carried out using Microsoft Power BI, a powerful business intelligence tool that enables interactive dashboards and real-time data exploration. The project involves data cleaning, transformation, and the application of analytical techniques to convert raw data into useful information.

The developed dashboard provides a comprehensive view of key metrics such as total sales, customer distribution, popular product categories, and preferred payment methods. These visualizations make it easier to interpret complex data and support informed decision-making.

Overall, this project demonstrates how data analytics and visualization tools can be effectively used to analyse consumer shopping trends, identify patterns, and generate actionable insights that are valuable for businesses in improving performance and achieving growth.

2) SOURCE OF DATASET

The dataset used in this project is titled **"Consumer_shopping_Trends_2026"**, which contains structured data representing consumer purchasing behaviour. This dataset has been used for academic and analytical purposes to study trends, patterns, and insights related to shopping activities.

The dataset includes multiple attributes that describe different aspects of consumer behaviour,

such as:

- Customer demographics (Age, Gender, Location)
- Product categories
- Purchase amount and transaction details
- Payment methods
- Purchase frequency and customer activity

The data is assumed to be collected from retail systems and e-commerce platforms, where customer transactions are recorded and stored for analysis. Such datasets are commonly available on public platforms like Kaggle or can be generated synthetically for learning and project purposes.

For this project, the dataset was imported into Microsoft Power BI, where it was processed, cleaned, and transformed before performing analysis. The tool enabled efficient handling of large datasets and supported the creation of interactive visualizations.

The dataset is suitable for analysing consumer shopping trends as it provides sufficient information to study customer preferences, spending habits, and market behaviour. This makes it highly useful for deriving insights that can support business decisions and strategic planning.

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Name: Consumer_Shoppin...

Structure | Relationships | Calculations | Calendars

Manage relationships | New measure | Quick measure | New column | New table | Mark as date table

awareness	time_pressure_level	gender	city_tier	shopping_preference	total_monthly_spend	Age Group	Tech Level	Digital Adoption %
3	1	Male	Tier 3	Store	1862386	20-34	Low Tech	0.496316307095411
8	6	Male	Tier 3	Store	4238596	Gen Z	Medium Tech	0.496316307095411
5	5	Male	Tier 3	Store	5860570	20-34	Low Tech	0.496316307095411
10	1	Male	Tier 3	Store	2089255	20-34	Low Tech	0.496316307095411
3	9	Male	Tier 3	Store	2710864	Gen Z	High Tech	0.496316307095411
5	2	Male	Tier 3	Store	3349384	40-54	High Tech	0.496316307095411
1	3	Male	Tier 3	Store	1810974	40-54	Low Tech	0.496316307095411
3	2	Male	Tier 3	Store	2635883	20-34	Medium Tech	0.496316307095411
8	8	Male	Tier 3	Store	747934	Boomers	Medium Tech	0.496316307095411
6	5	Male	Tier 3	Store	1255869	Gen Z	High Tech	0.496316307095411
8	7	Male	Tier 3	Store	1530654	20-34	Low Tech	0.496316307095411
2	8	Male	Tier 3	Store	760204	40-54	Low Tech	0.496316307095411
2	1	Male	Tier 3	Store	2976925	20-34	High Tech	0.496316307095411
1	3	Male	Tier 3	Store	4780812	Boomers	Medium Tech	0.496316307095411
2	4	Male	Tier 3	Store	4012382	Boomers	Medium Tech	0.496316307095411
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2	10	Male	Tier 3	Store	1208226	20-34	High Tech	0.496316307095411
1	3	Male	Tier 3	Store	2786788	40-54	Low Tech	0.496316307095411
9	6	Male	Tier 3	Store	2787287	Boomers	Low Tech	0.496316307095411
7	7	Male	Tier 3	Store	1060724	20-34	Low Tech	0.496316307095411
4	9	Male	Tier 3	Store	2316062	40-54	Medium Tech	0.496316307095411
4	8	Male	Tier 3	Store	4472028	40-54	Medium Tech	0.496316307095411

Data

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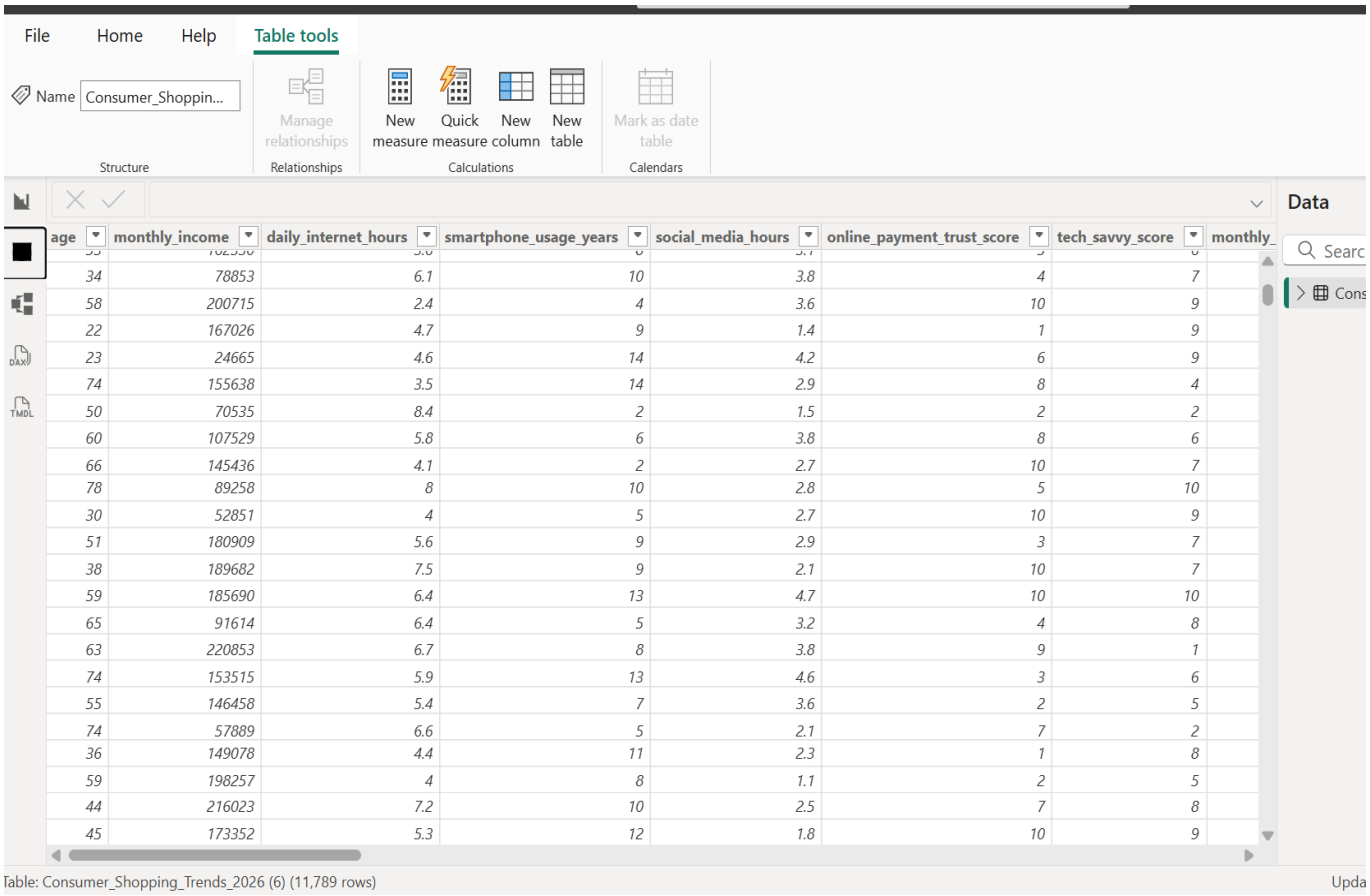
Consumer_Shopping_Trends_202...

Table: Consumer_Shopping_Trends_2026 (6) (11,789 rows)

Update available (click to download)



age	monthly_income	daily_internet_hours	smartphone_usage_years	social_media_hours	online_payment_trust_score	tech_savvy_score	monthly_
34	78853	6.1	10	3.8	4	7	
58	200715	2.4	4	3.6	10	9	
22	167026	4.7	9	1.4	1	9	
23	24665	4.6	14	4.2	6	9	
74	155638	3.5	14	2.9	8	4	
50	70535	8.4	2	1.5	2	2	
60	107529	5.8	6	3.8	8	6	
66	145436	4.1	2	2.7	10	7	
78	89258	8	10	2.8	5	10	
30	52851	4	5	2.7	10	9	
51	180909	5.6	9	2.9	3	7	
38	189682	7.5	9	2.1	10	7	
59	185690	6.4	13	4.7	10	10	
65	91614	6.4	5	3.2	4	8	
63	220853	6.7	8	3.8	9	1	
74	153515	5.9	13	4.6	3	6	
55	146458	5.4	7	3.6	2	5	
74	57889	6.6	5	2.1	7	2	
36	149078	4.4	11	2.3	1	8	
59	198257	4	8	1.1	2	5	
44	216023	7.2	10	2.5	7	8	
45	173352	5.3	12	1.8	10	9	

3) ANALYSIS ON DATASET

4.1 Customer Demographics Analysis

i. Introduction

Customer demographics analysis focuses on understanding the characteristics of consumers such as age, gender, and location. This analysis helps in identifying the target audience and understanding which customer groups contribute the most to overall sales.

ii. General Description

The dataset *Consumer_shopping_Trends_2026* includes demographic details that allow segmentation of customers into different groups. By analysing these attributes, patterns related to purchasing behaviour across various demographic segments can be identified.

iii. Specific Requirements, Functions and Formulas

The following functions and measures were used in Microsoft Power BI:

- **Total Customers** = COUNT (Customer ID)
- **Average Age** = AVERAGE (Age)
- **Total Sales by Gender** = SUM (Purchase Amount)
- Grouping customers into **Age Categories** using calculated columns

iv. Analysis Results

- The majority of customers fall within the **25–35 age group**, indicating a young and active consumer base.
- Both **male and female customers** contribute significantly, with slight variation depending on product categories.
- Customers from **urban areas** show higher purchasing activity compared to rural areas.
- Younger customers tend to make more frequent purchases, while older customers spend more per transaction.

v. Visualization

The following visualizations were created in Microsoft Power BI:

- Pie Chart for Gender Distribution
- Bar Chart for Age Group Analysis
- Map or Region Chart for Location-based Sales

4.2 Product Category Analysis

i. Introduction

Product category analysis examines the performance of different product categories in terms of sales and customer demand. This helps in identifying which products are most popular among consumers.

ii. General Description

The dataset contains various product categories such as electronics, fashion, groceries, and others. This analysis focuses on comparing these categories to determine their contribution to total revenue.

iii. Specific Requirements, Functions and Formulas

The following measures were used:

- **Total Sales by Category** = SUM (Purchase Amount)
- **Category-wise Customer Count** = COUNT (Customer ID)
- **Grouping data using Product Category field**

iv. Analysis Results

- **Electronics and Fashion categories** generate the highest sales revenue.
- Some categories show **seasonal trends**, indicating fluctuating demand over time.
- Essential goods such as groceries have **consistent but moderate sales**.
- High-value categories contribute significantly to overall profit despite fewer transactions.

v. Visualization

Visualizations used include:

- Bar Chart showing Sales by Category
- Line Chart showing Category Trends over Time
- Donut Chart showing Percentage Contribution

4.3 Payment Method Analysis

i. Introduction

Payment method analysis focuses on understanding how customers prefer to complete their transactions. This provides insights into the adoption of digital payment systems.

ii. General Description

The dataset includes different payment methods such as cash, credit card, debit card, and digital wallets. This analysis compares these methods to identify customer preferences.

iii. Specific Requirements, Functions and Formulas

The following calculations were used:

- **Transaction Count by Payment Method** = COUNT (Payment Method)
- **Percentage Contribution** using DAX formulas
- **Total Sales by Payment Type** = SUM (Purchase Amount)

iv. Analysis Results

- **Digital payment methods** (UPI, cards, wallets) are more widely used than cash.
- **Credit and debit cards** are preferred for high-value transactions.
- Cash payments are still used but are decreasing compared to digital methods.
- Frequent shoppers tend to rely more on digital payment options.

v. Visualization

The following visuals were created:

- Donut Chart for Payment Method Distribution
- Stacked Bar Chart for Payment vs Sales
- Pie Chart showing Percentage Usage

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Consumer Shopping Trends 2026
Total Revenue
2bn
Avg Online Spend
74.55K
Total_Customers
12K
Tech Savvy Index
5.53
city_tier
Deselect all
Tier 1
Tier 2
Tier 3
Gender
Deselect all
Female
Male
Other
Age Group
Deselect all
20-34
40-54
Boomers
Gen Z
The Revenue Gap
Average of avg_store_spend
Avg Online Spend
Average of avg_store_spend and Avg Online Spend
Preference Distribution
Sum of Digital Adoption %
Tier 1
Tier 3
Tier 2
city_tier
PREFERENCE SPLIT
0bn (5.82%)
2bn (92.24%)
shopping_prefer...
Store
Online
Hybrid
Count of monthly_income by gender
3.89K (33.01%)
3.97K (33.64%)
3.93K (33.34%)
gender
Male
Female
Other
Sum of delivery_fee_sensitivity and Sum of discount_sensitivity by Tech Level
Sum of delivery_fee_sensitivity
Sum of discount_s...
Medium Tech
High Tech
Low Tech
Tech Level
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Add data fields here
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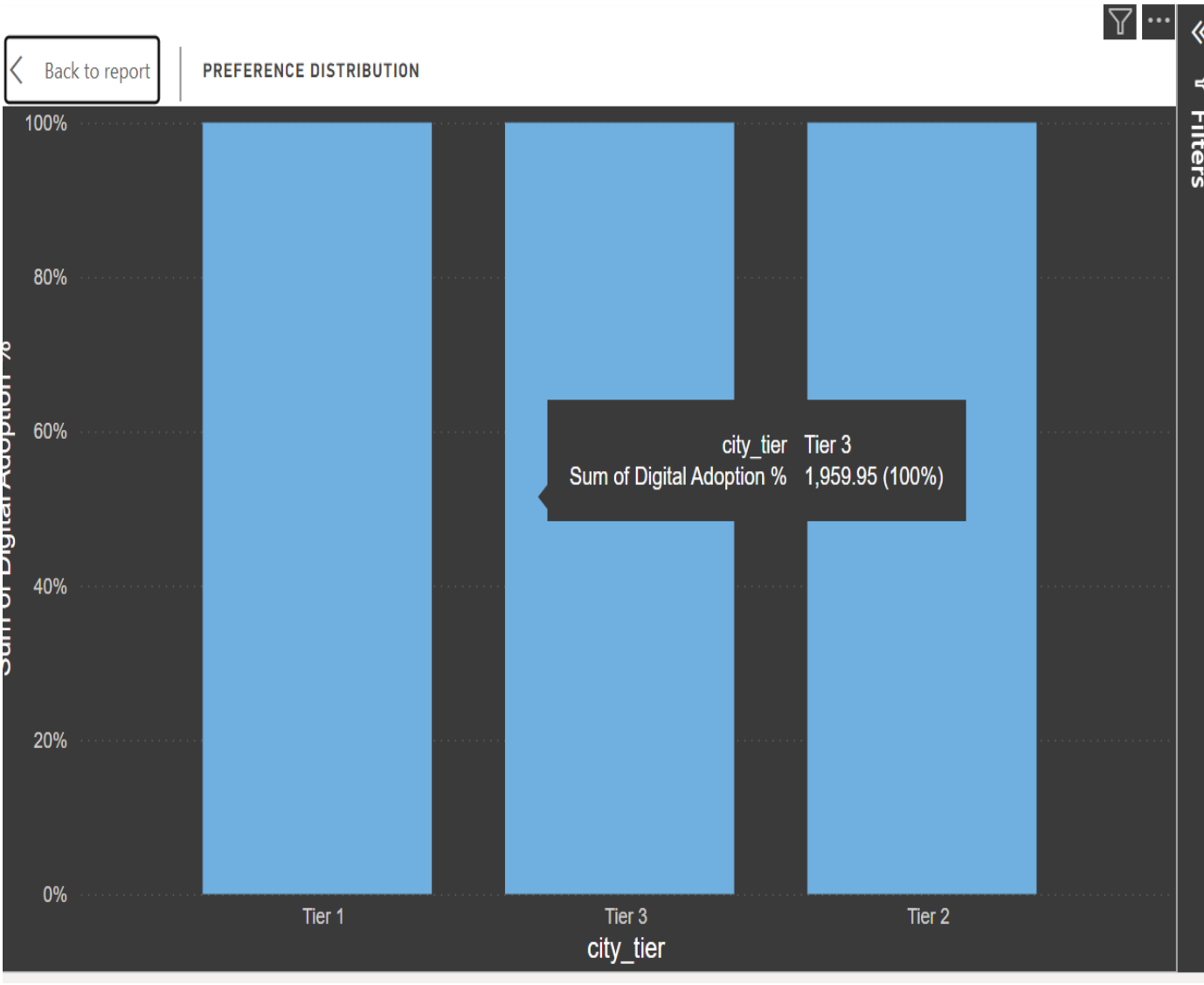
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PREFERENCE DISTRIBUTION
Filters
100%
80%
60%
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20%
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Sum of Digital Adoption %
city_tier Tier 3
Sum of Digital Adoption % 1,959.95 (100%)
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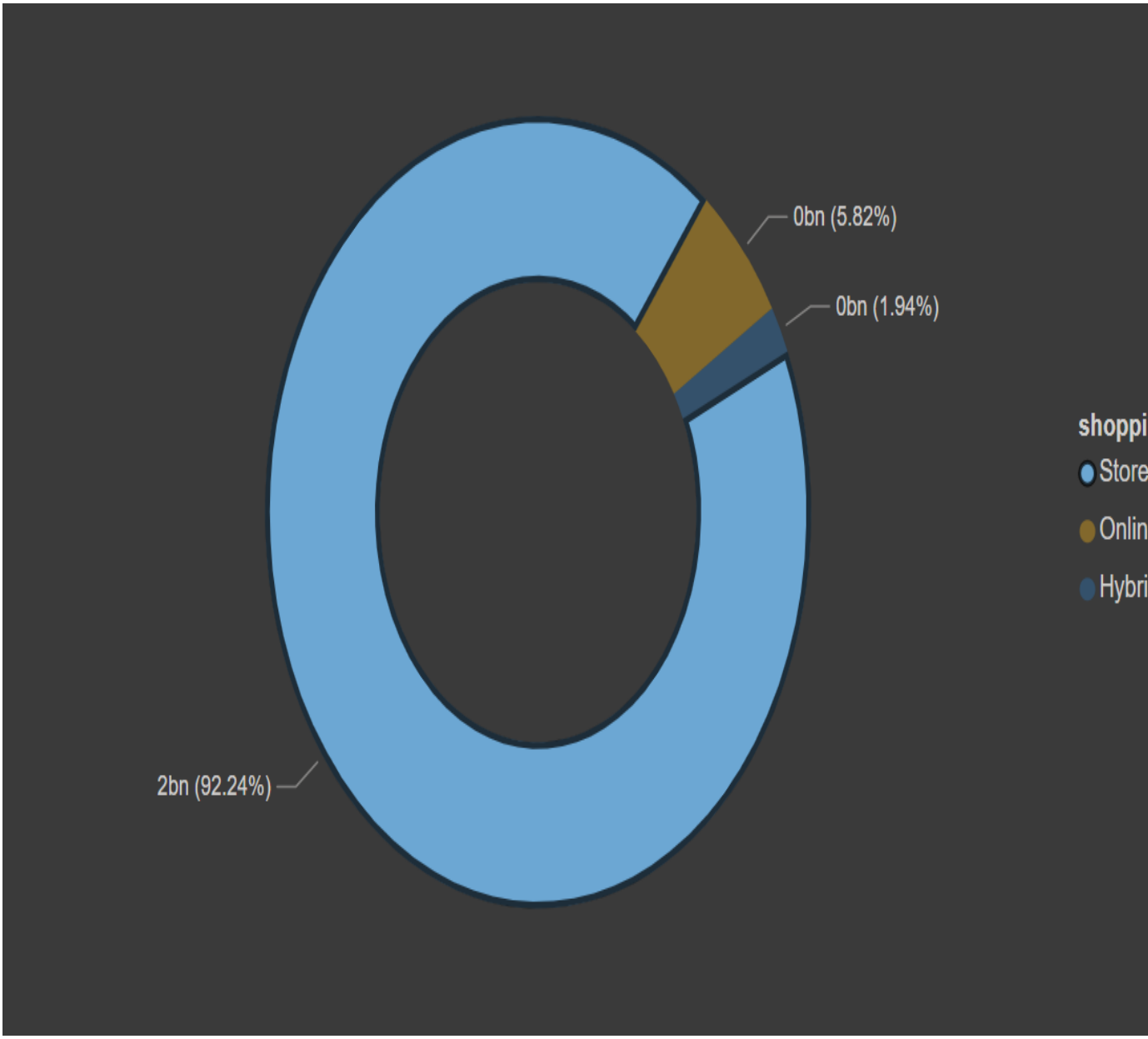
Tier 2
city_tier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2bn (92.24%)
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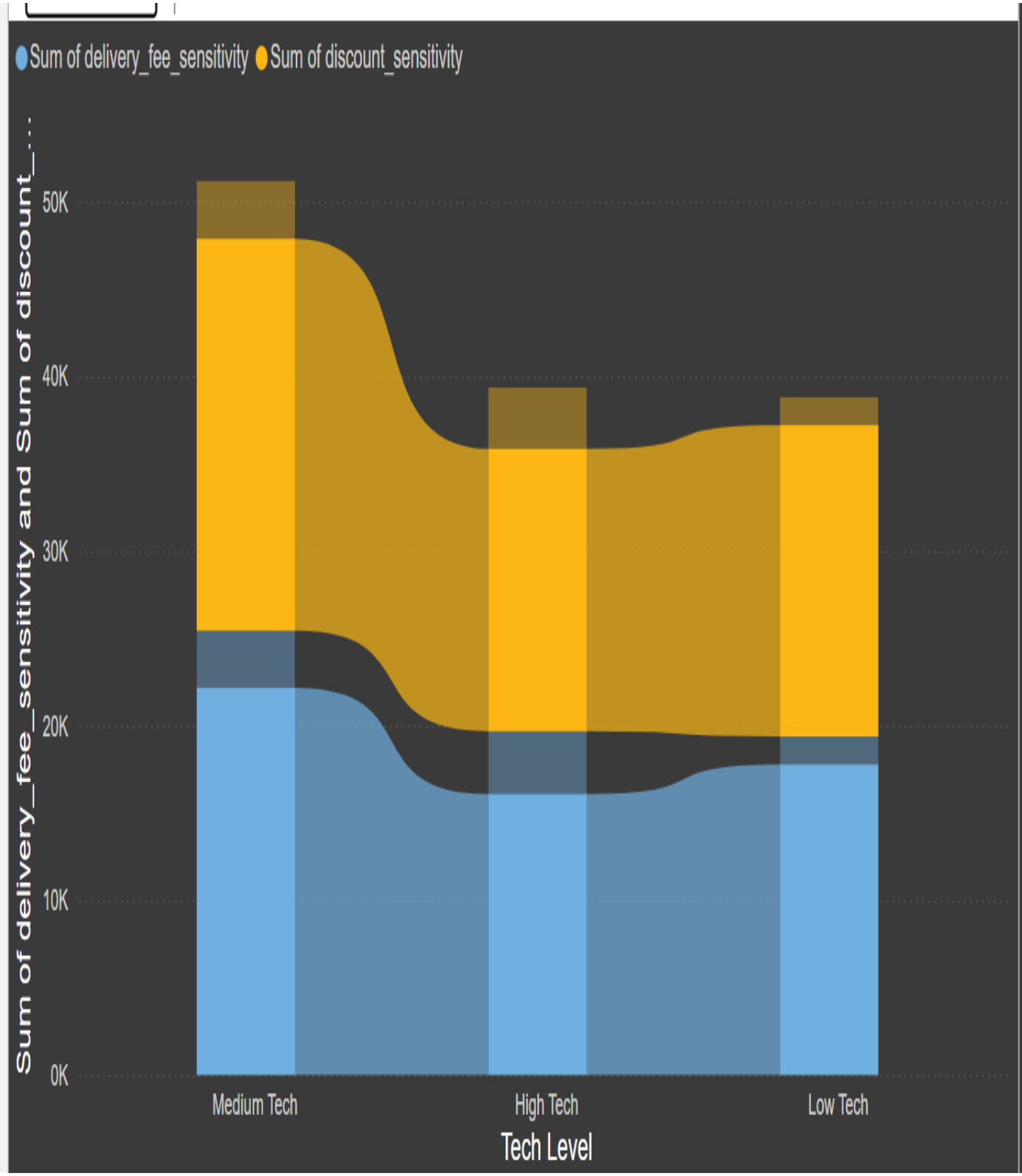
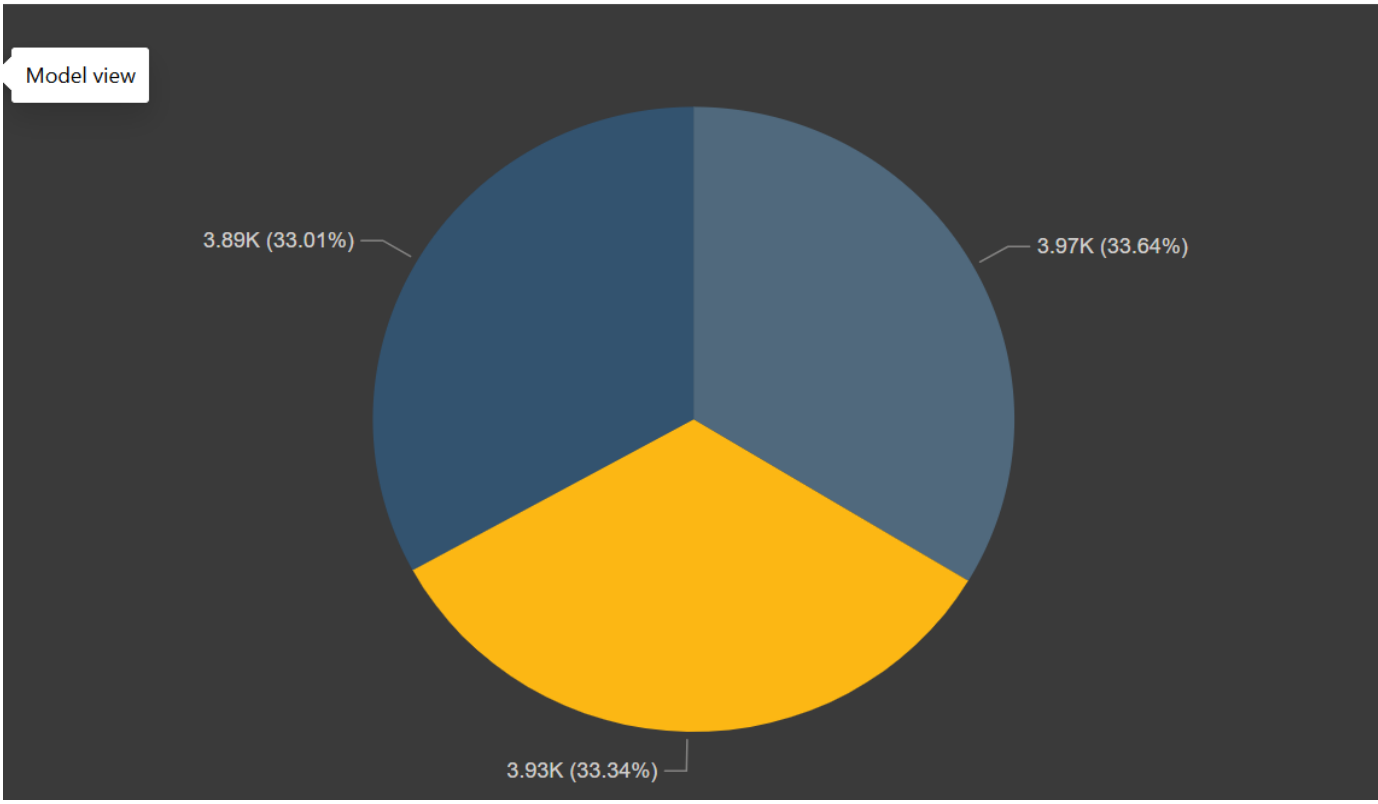
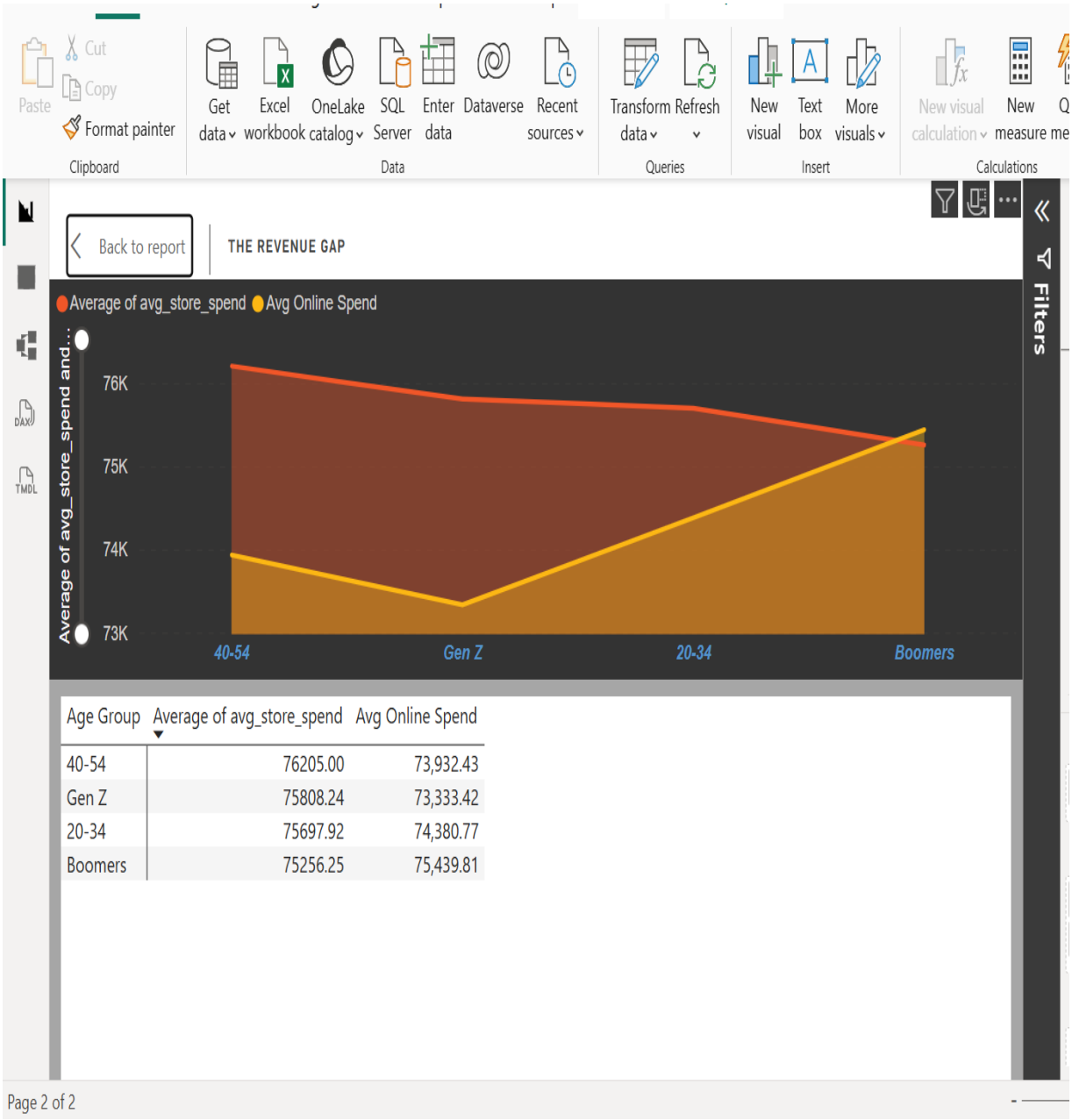
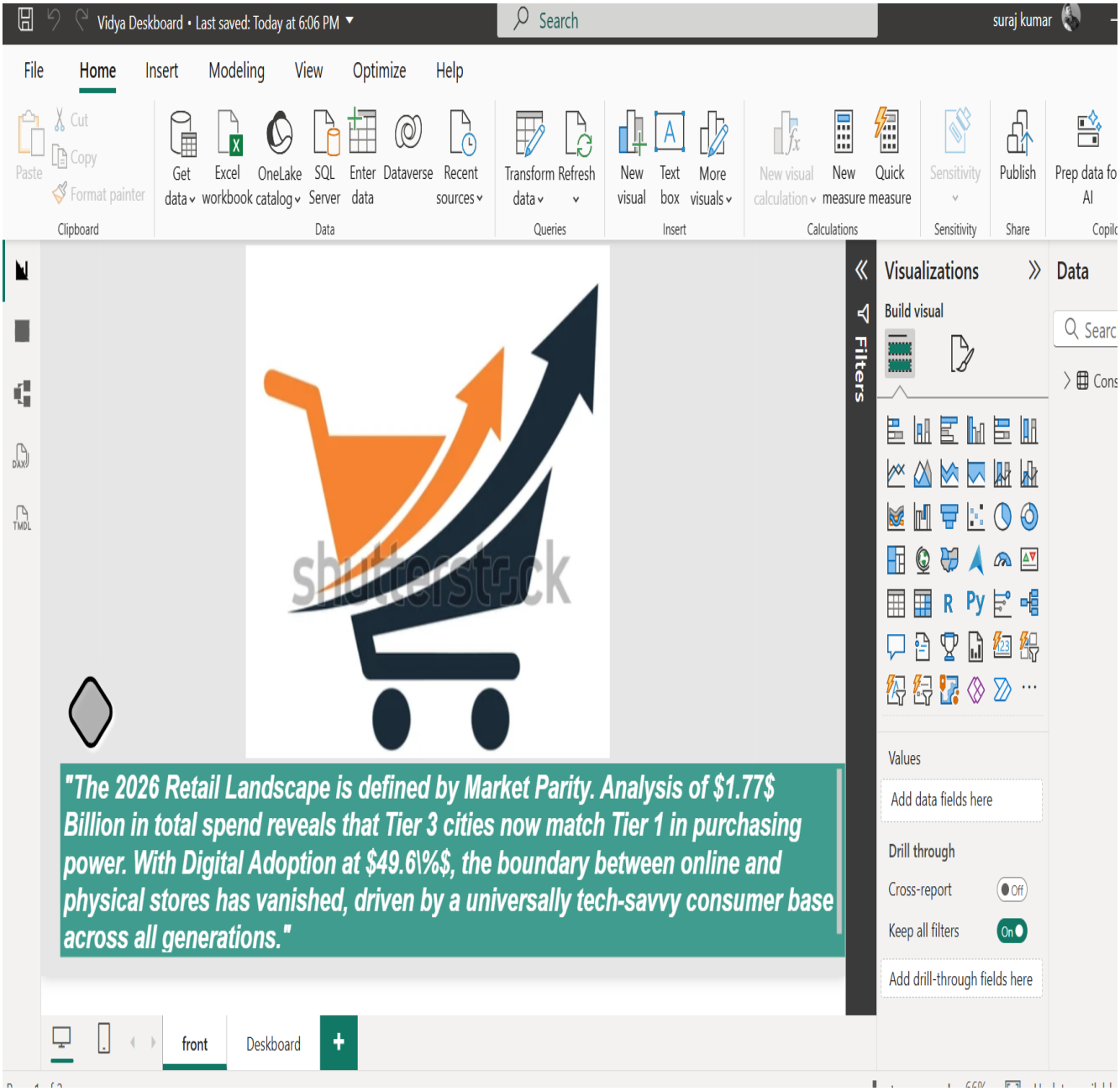

Sum of delivery_fee_sensitivity
Sum of discount_sensitivity
Sum of delivery_fee_sensitivity and Sum of discount_...
50K
40K
30K
20K
10K
0K
Medium Tech
High Tech
Low Tech
Tech Level

Back to report | **COUNT OF MONTHLY_INCOME** BY GENDER





Age Group	Average of avg_store_spend	Avg Online Spend
40-54	76205.00	73,932.43
Gen Z	75808.24	73,333.42
20-34	75697.92	74,380.77
Boomers	75256.25	75,439.81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OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data fo AI
Copilo
Filters
Visualizations
Build visual
Data
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
"The 2026 Retail Landscape is defined by Market Parity. Analysis of 1.77 Billion in total spend reveals that Tier 3 cities now match Tier 1 in purchasing power. With Digital Adoption at 49.6%, the boundary between online and physical stores has vanished, driven by a universally tech-savvy consumer base across all generations."
front
Deskboard

5) CONCLUSION

The project **"Consumer Shopping Trends Analysis 2026"** successfully explored and analysed consumer purchasing behaviour using the dataset *Consumer_shopping_Trends_2026*. By applying data cleaning, transformation, and visualization techniques, meaningful insights were extracted to understand customer preferences and market trends.

The analysis revealed several important findings. Customer demographics showed that the majority of consumers belong to younger age groups, with active participation from both male and female customers. Product category analysis highlighted that certain categories, such as electronics and fashion, contribute significantly to overall revenue, while essential goods maintain consistent demand. Additionally, payment method analysis indicated a clear shift toward digital transactions, reflecting the growing adoption of modern payment systems.

The use of Microsoft Power BI played a crucial role in transforming raw data into interactive and insightful visualizations. The developed dashboard enabled easy interpretation of complex data and supported effective decision-making through graphical representations.

Overall, this project demonstrates the importance of data analytics in understanding consumer behaviour and improving business strategies. The insights derived from this analysis can help organizations enhance customer targeting, optimize product offerings, and adopt suitable payment solutions to improve customer satisfaction and profitability.

In conclusion, the project highlights how business intelligence tools and data-driven approaches can be effectively utilized to gain valuable insights and support strategic growth in the retail and e-commerce sectors.

6) FUTURE SCOPE

The project **"Consumer Shopping Trends Analysis 2026"** provides valuable insights into consumer behaviour using data analytics and visualization techniques. However, there are several opportunities to further enhance and expand this project in the future.

One potential improvement is the **integration of real-time data** from live retail or e-commerce systems. This would allow businesses to monitor customer behaviour continuously and make timely decisions based on current trends rather than historical data alone.

Another important scope is the **application of machine learning techniques**. Predictive models can be developed to forecast customer purchasing behaviour, identify potential high-value customers, and recommend products based on past transactions. This would significantly improve personalization and customer engagement.

The project can also be extended by incorporating **larger and more diverse datasets**, including data from multiple regions, platforms, and time periods. This would provide deeper insights and improve the accuracy of analysis.

Further enhancement can be achieved by integrating the dashboard developed in Microsoft Power BI with other tools and platforms, enabling better scalability and collaboration across teams.

Additionally, the inclusion of **customer feedback and sentiment analysis** from reviews and social media platforms can provide a more comprehensive understanding of consumer preferences and satisfaction levels.

Another future direction is the development of **automated reporting systems**, where dashboards update dynamically and generate reports without manual intervention.

Finally, advanced features such as **mobile dashboard accessibility**, **interactive drill-down reports**, and **AI-powered insights** can be added to improve usability and user experience.

In conclusion, the project has significant potential for expansion by incorporating advanced analytics, real-time processing, and intelligent systems, making it more powerful and applicable in real-world business environments.

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Data Set: https://www.kaggle.com/datasets/sohaibdevv/consumer-shopping-behavior-and-preference-study-2026